36.2: Different mechanisms transport substances over short or long distances

Given the diversity of substances that move through plants and the great range of distances and barriers over which such substances must be transported, it is not surprising that plants employ a variety of transport processes. Before examining these processes, however, we'll look at the two major pathways of transport: the apoplast and the symplast.

The Apoplast and Symplast: Transport Continuums

Plant tissues have two major compartments—the apoplast and the symplast. The **apoplast** consists of everything external to the plasma membranes of living cells and includes cell walls, extracellular spaces, and the interior of dead cells such as vessel elements and tracheids (see Figure 35.10). The **symplast** consists of the entire mass of cytosol of all the living cells in a plant, as well as the plasmodesmata, the cytoplasmic channels that interconnect them.

The compartmental structure of plants provides three routes for transport within a plant tissue or organ: the apoplastic, symplastic, and transmembrane routes **(Figure 36.5)**. In the *apoplastic route*, water and solutes (dissolved chemicals) move along the continuum of cell walls and extracellular spaces. In the *symplastic route*, water and solutes move along the continuum of cytosol. This route requires substances to cross a plasma membrane once, when they first enter the plant. After entering one cell, substances can move from cell to cell via plasmodesmata. In the *transmembrane route*, water and solutes move out of one cell, across the cell wall, and into the neighboring cell, which may pass them to the next cell in the same way. The transmembrane route requires repeated crossings of plasma membranes as substances exit one cell and enter the next. These three routes are not mutually exclusive, and some substances may use more than one route to varying degrees.

Figure 36.5

Cell compartments and routes for short-distance transport.

Apoplastic route transport goes through cell walls. Symplastic route transport goes through center of cytosol and plasmodesma. Transmembrane route transport goes through the cell walls as well as and cytosol.

Some substances may use more than one transport route.

Cell compartments shown where the apoplast is the continuum of walls and extracellular spaces and the symplast is the continuum of cytosol connected by Plasmodesmata (channels that act as cytoplasmic bridges between plant cells).

Short-Distance Transport of Solutes Across Plasma Membranes

In plants, as in any organism, the selective permeability of the plasma membrane controls the short-distance movement of substances into and out of cells (see Concept 7.2). Both active and passive transport mechanisms occur in plants, and plant cell membranes are equipped with the same *general* types of pumps and transport proteins (channel proteins, carrier proteins, and cotransporters) that function in other cells. There are, however, *specific* differences between the membrane transport processes of plant and animal cells. In this section, we'll focus on some of those differences.

Unlike in animal cells, hydrogen ions (H+) rather than sodium ions (Na+) play the primary role in basic transport processes in plant cells. For example, in plant cells the membrane potential (the voltage across the membrane) is established mainly through the pumping of H+ by proton pumps **(Figure 36.6a)** rather than the pumping of Na+ by sodium-potassium pumps. Also, H+ is most often cotransported in plants, whereas Na+ is typically cotransported in animals. During cotransport, plant cells use the energy in the H+ gradient and membrane potential to drive the active transport of many different solutes. For instance, cotransport with H+ is responsible for absorption of neutral solutes, such as the sugar sucrose, by phloem cells and other plant cells. An H+/sucrose cotransporter couples movement of sucrose against its concentration gradient with movement of H+ down its electrochemical gradient **(Figure 36.6b)**. Cotransport with H+ also facilitates movement of ions, as in the uptake of nitrate (NO3−) by root cells **(Figure 36.6c)**.

Figure 36.6

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Solute transport across plant cell plasma membranes.
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The process of solute transport in four steps.

a. H plus and membrane potential. The plasma membranes of plant cells use A T P dependent proton pumps to pump H plus out of the cell. These pumps contribute to the membrane potential and the establishment of a p H gradient across the membrane. These two forms of potential energy can drive the transport of solutes.

b. H plus and cotransport of neutral solutes. Neutral solutes such as sugars can be loaded into plant cells by cotransport with H plus ions. H plus slash sucrose cotransporters, for example, play a key role in loading sugar into the phloem prior to sugar transport throughout the plant.

c. H plus and cotransport of ions. Cotransport mechanisms involving H plus also participate in regulating ion fluxes into and out of cells. For example, H plus slash N O 3 superscript minus cotransporters in the plasma membranes of root cells are important for the uptake of N O 3 superscript minus by plant roots.

d. Ion channels. Plant ion channels open and close in response to voltage, stretching of the membrane, and chemical factors. When open, ion channels allow specific ions to diffuse across membranes. For example, a K plus ion channel is involved in the release of K plus from guard cells when stomata close.

Assume that a plant cell has all four of the plasma membrane transport proteins shown above and that you have a specific inhibitor for each protein. Predict the effect of each inhibitor on the cell's membrane potential.

*For suggested answer, see*Appendix A.**

The membranes of plant cells also have ion channels that allow only certain ions to pass **(Figure 36.6d)**. As in animal cells, most channels are gated, opening or closing in response to stimuli such as chemicals, pressure, or voltage. Later in this chapter, we'll discuss how potassium ion channels in guard cells function in opening and closing stomata. Ion channels are also involved in producing electrical signals analogous to the action potentials of animals (see Concept 48.2). However, these signals are 1,000 times slower and employ Ca2+-activated anion channels rather than the sodium ion channels used by animal cells.

Short-Distance Transport of Water Across Plasma Membranes

The absorption or loss of water by a cell occurs by ****osmosis**, the diffusion of free water—water that is not bound to solutes or surfaces—across a membrane (see Figure 7.12). The physical property that predicts the direction in which water will flow is called **water potential**, a quantity that includes the effects of solute concentration and physical pressure. Free water moves from regions of higher water potential to regions of lower water potential if there is no barrier to its flow. The word *potential* in the term *water potential* refers to water's potential energy—water's capacity to perform work when it moves from a region of higher water potential to a region of lower water potential. For example, if a plant cell or seed is immersed in a solution that has a higher water potential, water will move into the cell or seed, causing it to expand. The expansion of plant cells and seeds can be a powerful force: The expansion of cells in tree roots can break concrete sidewalks, and the swelling of wet grain seeds within the holds of damaged ships can produce catastrophic hull failure and sink the ships. Given the strong forces generated by swelling seeds, it is interesting to consider whether water uptake by seeds is an active process. This question is examined in the **Scientific Skills Exercise**, which explores the effect of temperature on this process.

Scientific Skills Exercise

Calculating and Interpreting Temperature Coefficients

AP Science Practices 2, 4, 5, 6

Does the Initial Uptake of Water by Seeds Depend on Temperature? One way to answer this question is to soak seeds in water at different temperatures and measure the rate of water uptake at each temperature. The data can be used to calculate the temperature coefficient, Q10, the factor by which a physiological reaction (or process) rate increases when the temperature is raised by 10°C:

$$Q_{10}=(k_2k_1)^{10t_2-t_1}$$

where t_2 is the higher temperature (°C), t_1 is the lower temperature, k_2 is the reaction (or process) rate at t_2, and k_1 is the reaction (or process) rate at t_1. (If $t_2-t_1=10$, as here, the math is simplified.)

Q10 values may be used to make inferences about the physiological process under investigation. Chemical (metabolic) processes involving large-scale protein shape changes are highly dependent on temperature and have higher Q10 values, closer to 2 or 3. In contrast, many, but not all, physical parameters are relatively independent of temperature and have Q10 values closer to 1. For example, the Q10 of the change in the viscosity of water is 1.2–1.3. In this exercise, you will calculate Q10 using data from radish seeds (*Raphanus sativum*) to assess whether the initial uptake of water by seeds is more likely to be a physical or a chemical process.

How the Experiment Was Done Samples of radish seeds were weighed and placed in water at four different temperatures. After 30 minutes, the seeds were removed, blotted dry, and reweighed. The researchers then calculated the percent increase in mass due to water uptake for each sample.

Data from the Experiment

A table illustrates temperature and percent increase in mass due to water uptake after 30 minutes.

Data from J. D. Murphy and D. L. Noland, Temperature effects on seed imbibition and leakage mediated by viscosity and membranes, *Plant Physiology* 69:428–431 (1982).

Temperature	Percent increase in mass due to water uptake after 30 minutes
5 degrees Celsius	18.5
15 degrees Celsius	26.0
25 degrees Celsius	31.0

35 degrees Celsius	36.2

INTERPRET THE DATA

- Based on the data, does the initial uptake of water by radish seeds vary with temperature? What is the relationship between temperature and water uptake?
-
  - Using the data for 35°C and 25°C, calculate Q10 for water uptake by radish seeds. Repeat the calculation using the data for 25°C and 15°C and the data for 15°C and 5°C.
- What is the average Q10?
- Do your results imply that the uptake of water by radish seeds is mainly a physical process or a chemical (metabolic) process?
- Given that the Q10 for the change in the viscosity of water is 1.2–1.3, could the slight temperature dependence of water uptake by seeds be a reflection of the slight temperature dependence of the viscosity of water?
- Besides temperature, what other independent variables could you alter to test whether radish seed swelling is essentially a physical process or a chemical process?
- Would you expect plant growth to have a Q10 closer to 1 or 3? Why?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Water potential is abbreviated by the Greek letter Ψ (psi, pronounced “sigh”). Plant biologists measure Ψ in a unit of pressure called a **megapascal** (abbreviated MPa). By definition, the Ψ of pure water in a container open to the atmosphere under standard conditions (at sea level and at room temperature) is 0 MPa. One MPa is equal to about 10 times atmospheric pressure at sea level. The internal pressure of a living plant cell due to the osmotic uptake of water is approximately 0.5 MPa, about twice the air pressure inside an inflated car tire.

How Solutes and Pressure Affect Water Potential

Solute concentration and physical pressure are the major determinants of water potential in hydrated plants, as expressed in the water potential equation:

$$\Psi = \Psi S + \Psi P$$

where Ψ is the water potential, ΨS is the solute potential (osmotic potential), and ΨP is the pressure potential. The **solute potential** ΨS of a solution is directly proportional to its molarity. Solute potential is also called *osmotic potential* because solutes affect the direction of osmosis. The solutes in plants are typically mineral ions and sugars. By definition, the ΨS of pure water is 0. When solutes are added, they

bind water molecules. As a result, there are fewer free water molecules, reducing the capacity of the water to move and do work. In this way, an increase in solute concentration has a negative effect on water potential, which is why the ΨS of a solution is always expressed as a negative number. For example, a 0.1 M solution of a sugar has a ΨS of −0.23 MPa. As the solute concentration increases, ΨS will become more negative.

Pressure potential ΨP is the physical pressure on a solution. Unlike ΨS, ΨP can be positive or negative relative to atmospheric pressure. For example, when a solution is being withdrawn by a syringe, it is under negative pressure; when it is being expelled from a syringe, it is under positive pressure. The water in living cells is usually under positive pressure due to the osmotic uptake of water. Specifically, the **protoplast** (the living part of the cell, which also includes the plasma membrane) presses against the cell wall, creating what is known as **turgor pressure**. This pushing effect of internal pressure, much like the air in an inflated tire, is critical for plant function because it helps maintain the stiffness of plant tissues and also serves as the driving force for cell elongation. Conversely, the water in the hollow nonliving xylem cells (tracheids and vessel elements) of a plant is often under a negative pressure potential (tension) of less than −2 MPa.

As you learn to apply the water potential equation, keep in mind the key point: *Water moves from regions of higher water potential to regions of lower water potential.*

Water Movement Across Plant Cell Membranes

Now let's consider how water potential affects absorption and loss of water by a living plant cell. First, imagine a cell that is ****flaccid** (limp) as a result of losing water. The cell has a ΨP of 0 MPa. Suppose this flaccid cell is bathed in a solution of higher solute concentration (more negative solute potential) than the cell itself **(Figure 36.7a)**. Since the external solution has the lower (more negative) water potential, water diffuses out of the cell. The cell's protoplast undergoes ***plasmolysis—that is, it shrinks and pulls away from the cell wall. If we place the same flaccid cell in pure water (Ψ=0 MPa)***(Figure 36.7b)**, the cell, because it contains solutes, has a lower water potential than the water, and water enters the cell by osmosis. The contents of the cell begin to swell and press the plasma membrane against the cell wall. The partially elastic wall, exerting turgor pressure, confines the pressurized protoplast. When this pressure is enough to offset the tendency for water to enter because of the solutes in the cell, then ΨP and ΨS are equal, and Ψ=0. This matches the water potential of the extracellular environment—in this example, 0 MPa. A dynamic equilibrium has been reached, and there is no further *net* movement of water.

Figure 36.7

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Water relations in plant cells.
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Water relations in plant cells.

In these experiments, flaccid cells (cells in which the protoplast contacts the cell wall but lacks turgor pressure) are placed in two environments. The blue arrows indicate initial net water movement.

Part a. Initial flaccid cell enters environment with 0.4 molarity sucrose solution. Final plasmolyzed cell has the same water potential as its surroundings. Initial flaccid cell: Psi sub P = 0. Psi sub S = negative 0.7. Psi = Psi sub P + Psi sub S = negative 0.7 mega Pascal. Environment: 0.4 molarity sucrose solution. Psi sub P = 0. Psi sub S = negative 0.9. Psi = Psi sub P + Psi sub S = negative 0.9 mega Pascal. Water diffuses out of the cell since the external solution has lower water potential than inside the cell, and the protoplast shrinks and pulls away from the cell wall. Final plasmolyzed cell at osmotic equilibrium with its surroundings: Psi sub P = 0. Psi sub S = negative 0.9. Psi = Psi sub P + Psi sub S = negative 0.9 mega Pascal.

Part b. Initial flaccid cell enters environment with pure water. Final turgid cell has the same water potential as its surroundings. Initial flaccid cell: Psi sub P = 0. Psi sub S = negative 0.7. Psi = Psi sub P + Psi sub S = negative 0.7 mega Pascal. Environment: Pure water. Psi sub P = 0. Psi sub S = 0. Psi = Psi sub P + Psi sub S = 0 mega Pascal. Inside of the cell, with more solutes, has lower water potential than the outside of the cell and water flows into the cell by osmosis. Final turgid cell at osmotic equilibrium with its surroundings: Psi sub P = 0.7. Psi sub S = negative 0.7. Psi = Psi sub P + Psi sub S = 0 mega Pascal.

Video:

Plasmolysis

image

Play

image

Play

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In contrast to a flaccid cell, a walled cell with a greater solute concentration than its surroundings is ****turgid**, or very firm. When turgid cells in a nonwoody tissue push against each other, the tissue is stiffened. The effects of turgor loss are seen during **wilting**, when leaves and stems droop as a result of cells losing water.

A turgid plant appears to be healthy and vibrant. A wilted plant appears to be unhealthy and dying with drooping leaves and flowers.

Aquaporins: Facilitating Diffusion of Water

A difference in water potential determines the *direction* of water movement across membranes, but how do water molecules actually cross the membranes? Water molecules are small enough to diffuse across the phospholipid bilayer, even though the bilayer's interior is hydrophobic. However, their movement across biological membranes is too rapid to be explained by unaided diffusion. Transport proteins called ******aquaporins** (see Figure 7.10 and Concept 7.2) facilitate the transport of water molecules across plant cell plasma membranes. Aquaporin channels, which can open and close, affect the *rate* at which water moves osmotically across the membrane. Their permeability is decreased by increases in cytosolic Ca2+ or decreases in cytosolic pH.

Long-Distance Transport: The Role of Bulk Flow

Diffusion is an effective transport mechanism over the spatial scales typically found at the cellular level. However, diffusion is much too slow to function in long-distance transport within a plant. Although diffusion from one end of a cell to the other takes just seconds, diffusion from the roots to the top of a giant redwood would take several centuries. Instead, long-distance transport occurs through **bulk flow**, the movement of liquid in response to a pressure gradient. The bulk flow of material always occurs from higher to lower pressure. Unlike osmosis, bulk flow is independent of solute concentration.

Long-distance bulk flow occurs within specialized cells in the vascular tissue, namely, the tracheids and vessel elements of the xylem and the sieve-tube elements of the phloem. In leaves, the branching of veins ensures that no cell is more than a few cells away from the vascular tissue **(Figure 36.8)**.

Figure 36.8

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Venation in an aspen leaf.
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A magnified photograph shows clusters of individual leaf cells surrounded by leaf veins.

The finer and finer branching of leaf veins in eudicot leaves ensures that no leaf cell is far removed from the vascular system.

VISUAL SKILLS *In this leaf, what is the maximum number of cells any mesophyll cell is from a vein?*

*For suggested answer, see*Appendix A.**

The structures of the conducting cells of the xylem and phloem facilitate bulk flow. Mature tracheids and vessel elements are dead cells and therefore have no cytoplasm, and the cytoplasm of sieve-tube elements (also called sieve-tube members) is almost devoid of organelles (see Figure 35.10). If you have dealt with a partially clogged drain, you know that the volume of flow depends on the pipe's diameter. Clogs reduce the

effective diameter of the drainpipe. Such experiences help us understand how the structures of plant cells specialized for bulk flow fit their function. Like unclogging a drain, the absence or reduction of cytoplasm in a plant's "plumbing" facilitates bulk flow through the xylem and phloem. Bulk flow is also enhanced by the perforation plates at the ends of vessel elements and the porous **sieve plates** connecting sieve-tube elements.

Diffusion, active transport, and bulk flow act in concert to transport resources throughout the whole plant. For example, bulk flow due to a pressure difference is the mechanism of long-distance transport of sugars in the phloem, but active transport of sugar at the cellular level maintains this pressure difference. In the next three sections, we'll examine in more detail the transport of water and minerals from roots to shoots, the control of evaporation, and the transport of sugars.